

Battle of the SoCs



Akul Sabharwal is an electronics and communication engineer. He is currently working with Simmtronics Semiconductors Ltd

One are the days when mobile phones were considered a luxury. Nowadays, necessity is the best way to describe these. In the past, design and brand were the only things that people focused on before purchasing a handset. Changing trends and high demands from the consumer have led to major changes in the central processing unit (CPU). Your camera, screen resolution, applications and operating system (OS) can only improve if the CPU becomes more powerful. Smartphones have gone from single-core to dual-core to quad-core, and are now gradually moving towards octa-core. System on chip (SoC) is a very common term in the industry, but what is it exactly?

What is SoC

An SoC is an integrated circuit that integrates all components of an electronic system into a single chip. Rather than having to implement the CPU, graphics processing unit (GPU), etc individually, manufacturers can implement the SoC and reap its benefits.

Following is a comparison between two popular SoCs: Nvidia Tegra K1 and Qualcomm Snapdragon 801.

Nvidia's SoC features Kepler GPU, four Plus-1 Cortex A15 processors and dual ISP cores, among others, whereas Qualcomm

has managed to pack all of that, as well as wireless radios, like GPS, Wi-Fi and LTE, on a single chip. Both Snapdragon 801 and Tegra K1 use a smaller fabrication process (28nm) for the chip.

ARM, the backbone

ARM is a 32-bit microprocessor originally created by Acorn Computers in 1987. Since then, ARM has been the preferred microchip by any and all companies looking for a cheap and, more importantly, low-powered chip for portable devices. ARM uses a completely different architecture as compared to Intel and AMD, who have stuck with x86 architecture, and hence all three companies have happily co-existed.

Since ARM architecture is licensable, companies can either obtain an architectural licence to design their own customised CPU or simply choose from one of ARM's designs. Popular core designs from ARM include Cortex-A7, Cortex-A8, Cortex-A9 and Cortex-A15. Companies like Samsung and Apple simply choose one of these designs and package these, along with a GPU and other components, into their own branded SoCs. However, SoC manufacturers like Qualcomm prefer to design their own custom CPUs, rather than go with the crowd.

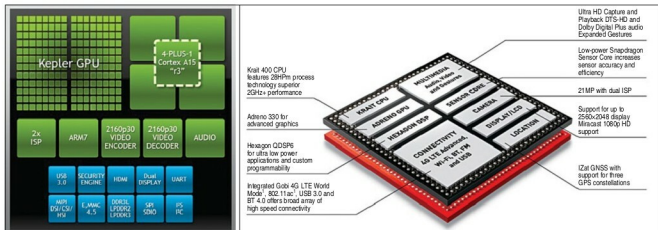


Fig. 1: Nvidia Tegra K1 SoC (left) and Qualcomm Snapdragon 801 SoC (right)